

Proposal details

Title of the project: Combining two-photon polymerisation and melt electron writing for printing multiscale vascular systems.

Duration of the project (in months): 12

Public summary

English public summary:

Replicating the complex hierarchical multiscale structure of vascular networks typically requires compromising on either printing speed or resolution, depending on the printing process. 3D Bioprinting harms living cells over the deposition time; therefore, a rapid printing process is preferred. However, a faster printing process puts more stress on cells, which should be minimised as well. A hybrid setup consisting of a faster micrometre scale process (MEW), together with a slower nanometre scale process (TPP) for precise sections, is proposed. The work plan comprises four intermediate challenges: a printable material analysis, adherence between MEW and TPP, the overall process and bioactivity.

Budget

Type of costs	Clarification	Costs in euros
<i>Personnel</i>	<i>PhD student, 1FTE, 11 months</i>	<i>€ 44.001</i>
<i>Project related goods</i>	<i>Bioinks and lab disposables</i>	<i>€ 5.000</i>
Total request from NWO		€ 49.001

Budget justification (0/150)

The 3D printers are readily available in the group, the material budget is solely spent on operational costs.

Project proposal

3D bioprinting (3DBP), an additive manufacturing process of tissue engineering, has advanced into an established field. The field has seen many variations of bioinks deposited using various printing processes, each with its pros and cons. One challenge in 3DBP is recreating the highly complex vascular networks of natural tissue; these networks range from centimetre to nanometre scale. This hierarchy is responsible for the localised delivery of oxygen and nutrients where required [1].

Replicating the complex hierarchical multiscale structure of natural vascular networks by 3D bioprinting requires compromising on either speed or resolution, depending on the chosen printing process. This is a common trade-off in 3D printing processes, where higher-resolution prints often inhibit fast deposition.

3DBP is sometimes referred to as 4D bioprinting, as time can be another factor to consider, given the living cells that are put into bioinks. This can provide opportunities, such as a delayed release of cell activity [2]. But mostly, the printing process harms living cells over time, and therefore, a rapid printing process is preferred. However, typically a faster printing process will put more stress on the cells, which should be minimised as well [1]. This would make a process that can print relatively quickly at high resolutions, without increasing the stress applied to printed cells, a beneficial addition to the 3D bioprinting field.

One way of achieving such results could be through a hybrid of two promising printing processes combined in a single printing setup. A faster micrometre scale process could print larger bulk parts, together with a slower nanometre scale process for precise sections, allowing for faster printing at a high resolution.

The first printing process to be implemented into such a combined setup, melt electron writing (MEW), applies the fundamentals of electrospinning and fused deposition modelling (FDM). A high-voltage electric field pulls bioink from the grounded nozzle to the charged collector bed, while the nozzle's movement is operated similarly to FDM [2]. MEW performs at a microscopic scale, defined here as the larger bulk scale. MEW has been applied to vascular printing setups before, creating scaffolds with locally tuneable mechanical properties. Some downsides of the MEW process include a limited assortment of materials it can deposit, little bioactivity and complex thermal control [3].

A second printing process, working at nanometre precision, is two-photon polymerisation (TPP). In TPP, a drop of resin is deposited onto the substrate and locally polymerised by a femtosecond-pulsed laser, after which the excess resin is removed. TPP has been applied to microvascular printing setups before with adjustable mechanical properties, and its prints integrate well in tissue and provide transport of nutrients and oxygen. The TPP process is most limited by the system's complexity and low production speed [3].

Just as MEW is similar to FDM in the underlying mechanics, TPPs' underlying principles are similar to vat polymerisation (VP). This is a printing process applying stereolithography onto a vat of photo resin, polymerising the top layer of resin onto the substrate, which then lowers after every polymerised layer, allowing the next layer to print onto the previous layer. FDM and VP have already been combined into a hybrid setup, allowing for multi-material prints and circumventing some of the weaknesses of either process [4].

To create a hybrid vascular printing setup with MEW and TPP, several intermediate steps will be required. These steps and their are composed into a work plan consisting of four Work Packages (WP):

WP1

First, an analysis of possible materials to print with should be performed. This part starts with a literature review on the optimal parameters and materials to deposit with for each printer, and is continued by an analysis of the differences between the printing processes. The goal is to determine the possibilities of depositing materials and structures, to hypothesise how the combined setup could be applied. There is no need to add cells at this stage yet. This part will consist of both qualitative and quantitative parts, creating a matrix for the optimal parameters and materials.

WP2

Next, the adherence between a MEW-printed layer and a TPP-printed layer will be determined and optimised. This optimisation will have to go both ways: a TPP layer printed onto a MEW layer, and a MEW layer onto a TPP layer. The adherence should be at the exact spot where deposited, should not deform the other layer, leave no inclusions between processes or other unwanted defects. This part will consist of both qualitative and quantitative parts, describing the pros and cons of different strategies and materials. This includes both deposition of similar materials and multimaterial deposition performed by the combination of processes.

WP3

Third, the overall process of the combined setup should be improved; the combined process of VP and FDM is described as “coating-curing-blowing-extrusion-curing” [4], requiring the material from the previous process to be cured and cleaned before continuing deposition with the next process. As a baseline, this process could be attempted to replicate for MEW and TPP. However, for the deposition time to decrease, other process strategies should be considered. Perhaps the deposited bioink could be polymerised at a later stage in the printing process for TPP, allowing for fewer swaps between processes. This would require the MEW-deposited material to allow stereolithography to pass through and the printed structure to dissipate excess material. Other possible optimisation strategies should be considered as well.

WP4

Finally, for the combined process to be considered a successful addition to the 3D bioprinting field, the process should allow for biocompatibility and control. The effect of the printing process in combination with the chosen materials on the cells will be studied and described. Then, by altering the process and setup parameters, the harm done to printed cells should be minimised while aiming to maintain similar adherence and process speed. The level of control possible on bioactivity will be documented at a set of points in time before, during and after printing.

Literature references

- [1] Mahdi Kazemi and Sepideh Maralbashi. "Advances in 3D bioprinting for medical application: opportunities and challenges". In: BioMedical Engineering OnLine 25 (2025). doi: 10 . 1186 / s12938 - 025 - 01498 - y. url: <https://doi.org/10.1186/s12938-025-01498-y>.
- [2] Yanping Zhang et al. "Rational Design and Functionalization of Melt Electrowritten 4D Scaffolds for Biomedical Applications". In: Nano-Micro Letters 18 (2026). doi: 10.1007/s40820-025-01986-9. url: <https://doi.org/10.1007/s40820-025-01986-9>.
- [3] Min Suk Kim, Yuri Choi, and Keel Yong Lee. "Three-Dimensional Printing and Bioprinting Strategies for Cardiovascular Constructs: From Printing Inks to Vascularization". In: Polymers 17 (2025), p. 2337. doi: 10.3390/polym17172337. url: <https://doi.org/10.3390/polym17172337>.
- [4] Hewu Sun et al. "Multi-material ceramic hybrid additive manufacturing based on vat photopolymerization and material extrusion compound process". In: Additive Manufacturing 97 (2025), p. 104627. doi: 10.1016/j.addma.2024.104627. url: <https://doi.org/10.1016/j.addma.2024.104627>.